

The cross-polytope disproves the conjectured random-section radius bound

Autonomous proof packet for arXiv:1601.02254

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Abstract

Question 1.1 of Giannopoulos–Hioni–Tsolomitis asks whether every volume-one isotropic convex body $K \subset \mathbb{R}^n$ has

$$R(K \cap F) \leq C \sqrt{n/k} \sqrt{n} L_K = C n L_K / \sqrt{k}$$

for a random codimension- k subspace F , with one universal constant. The answer is negative. For the volume-normalized cross-polytope, the sharp lower bound for the Gelfand widths of $\ell_1^n \hookrightarrow \ell_2^n$ forces an additional $\sqrt{1 + \log(n/k)}$ factor for *every* codimension- k subspace. Taking $k = \lfloor \sqrt{n} \rfloor$ contradicts the proposed estimate.

1 The question

The source asks for an absolute C such that, for every isotropic convex body $K \subset \mathbb{R}^n$ and every $1 \leq k \leq n-1$, a random $F \in G_{n,n-k}$ satisfies

$$R(K \cap F) \leq C \frac{n L_K}{\sqrt{k}}. \quad (1)$$

Here $R(A) = \sup\{\|x\|_2 : x \in A\}$.

Question 1.1. *There exists an absolute constant $\bar{c}_0 > 0$ with the following property: for every isotropic convex body K in $\mathbb{R}^n$ and for every $1 \leq k \leq n-1$, a random subspace $F \in G_{n,n-k}$ satisfies*

$$(1.4) \quad R(K \cap F) \leq \bar{c}_0 \sqrt{n/k} \sqrt{n} L_K.$$

2 The width input

For $0 \leq m < n$, define the m th Gelfand width by

$$d^m(B_1^n, \ell_2^n) := \inf_{A \in \mathbb{R}^{m \times n}} \sup\{\|x\|_2 : x \in B_1^n \cap \ker A\}.$$

The sharp classical estimate of Garnaev–Gluskin, in the convenient form of Foucart–Pajor–Rauhut–Ullrich [2, Theorem 1.1], says that there is a universal $c > 0$ such that

$$d^m(B_1^n, \ell_2^n) \geq c \min \left\{ 1, \sqrt{\frac{1 + \log(n/m)}{m}} \right\} \quad (1 \leq m < n). \quad (2)$$

Only this lower bound is needed below. Notice that it is an infimum over all kernels. Consequently, if F has codimension m and A is any rank- m map with $\ker A = F$, then

$$R(B_1^n \cap F) \geq d^m(B_1^n, \ell_2^n). \quad (3)$$

3 Normalization of the cross-polytope

Let

$$a_n := \left(\frac{n!}{2^n}\right)^{1/n}, \quad K_n := a_n B_1^n.$$

Since $|B_1^n| = 2^n/n!$, the body K_n has volume one. Sign and coordinate-permutation symmetries show that its barycenter is zero and its covariance matrix is a scalar multiple of the identity, so it is isotropic. We record the scalar exactly.

Lemma 1. *The isotropic constant of K_n satisfies*

$$L_{K_n}^2 = \frac{2a_n^2}{(n+1)(n+2)}.$$

Proof. On the positive-orthant simplex $\Delta_n = \{x_i \geq 0 : \sum_i x_i \leq 1\}$, the Dirichlet integral gives

$$\int_{\Delta_n} x_1^2 dx = \frac{2}{(n+2)!}.$$

After summing over the 2^n orthants and dividing by $|B_1^n| = 2^n/n!$, one obtains

$$\frac{1}{|B_1^n|} \int_{B_1^n} x_1^2 dx = \frac{2}{(n+1)(n+2)}.$$

The mixed second moments vanish by sign symmetry, and scaling by a_n proves the claim. $\square$

In particular,

$$\frac{nL_{K_n}}{\sqrt{k}} = \frac{a_n}{\sqrt{k}} \frac{\sqrt{2}n}{\sqrt{(n+1)(n+2)}} \leq \frac{\sqrt{2}a_n}{\sqrt{k}}. \quad (4)$$

4 Counterexample

Theorem 1. *There is no universal constant C for which (1) holds. More precisely, if $k_n = \lfloor \sqrt{n} \rfloor$, then for all sufficiently large n and every $F \in G_{n,n-k_n}$,*

$$\frac{R(K_n \cap F)}{nL_{K_n}/\sqrt{k_n}} \geq c' \sqrt{1 + \log(n/k_n)},$$

where $c' > 0$ is universal. Hence the ratio tends to infinity.

Proof. Homogeneity, (3), and (2) give, once n is large enough that the second branch of the minimum applies,

$$R(K_n \cap F) = a_n R(B_1^n \cap F) \geq ca_n \sqrt{\frac{1 + \log(n/k_n)}{k_n}}.$$

Dividing this inequality by (4) yields

$$\frac{R(K_n \cap F)}{nL_{K_n}/\sqrt{k_n}} \geq \frac{c}{\sqrt{2}} \sqrt{1 + \log(n/k_n)}.$$

Since $k_n = \lfloor \sqrt{n} \rfloor$, the last expression diverges. For any proposed constant C , choose n so large that it exceeds C . Then (1) fails for every codimension- k_n subspace, and therefore cannot hold for a random one. $\square$

5 Remarks on scope and novelty

The argument answers only Question 1.1 of the source; it does not address the separate question about the existence of a dimension-free sub-Gaussian direction. The width estimate predates the source question, but bounded searches of the run indexes and the arXiv-facing literature did not find an explicit application identifying the volume-normalized cross-polytope as a counterexample. The mathematical content of this packet is the short normalization-and-width deduction.

References

- [1] A. Giannopoulos, L. Hioni, and A. Tsolomitis, *Geometry of random sections of isotropic convex bodies*, arXiv:1601.02254v2 (2016).
- [2] S. Foucart, A. Pajor, H. Rauhut, and T. Ullrich, *The Gelfand widths of ℓ_p -balls for $0 < p \leq 1$* , J. Complexity 26 (2010), 629–640; arXiv:1002.0672v2.